DEVELOPER GUIDE · LOCAL ENVIRONMENT

Docker Desktop on Windows

Installation, configuration and day-to-day usage for the Spencer Denim E-Commerce Platform

Project	Spencer Denim E-Commerce Platform	Prepared by	TU2SL
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Audience	Developers, IT administrators	Platform	Windows 10 22H2 / 11 (64-bit)

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1. Purpose and scope

This guide explains how to install Docker Desktop on a Windows development PC and use it to run the Spencer Denim platform locally. The platform has a .NET 10 API, an Angular 21 storefront and admin panel, SQL Server 2025 Express, and a Caddy web server.

Docker runs each part of the system in an isolated container, so every developer's PC matches the production server (Hostinger VPS, Ubuntu 24.04) as closely as possible.

> **Note:** Parts C and D describe the project's Docker setup as designed. The exact file names and commands are confirmed in the project `README.md` once the project scaffold (task W0-09) is complete.

2. System requirements

Requirement	Minimum / recommended	Development PC (checked 16 Sep 2026)
Operating system	Windows 10 22H2 (build 19045+) or Windows 11, 64-bit, Pro/Enterprise	Windows 10 Pro, build 19045 ✓
CPU virtualization	Enabled in BIOS/UEFI	Enabled ✓
Memory (RAM)	8 GB minimum · 16 GB recommended	16 GB ✓
Free disk space	30 GB+ (images, database volumes)	C: 60 GB · D: 105 GB ✓
WSL 2 (Windows Subsystem for Linux)	Required	**Not installed** (Part A)

Administrator rights	Required *for installation only*	**Not available**: IT must install
Network	Access to Docker Hub and mcr.microsoft.com	Check company proxy/firewall
Already installed	.NET SDK 10, Node.js 24, Angular CLI 21, Git, GitHub CLI	All present ✓

3. Licensing

Docker Desktop is **free** for personal use, education, non-commercial open-source projects, and small businesses with **fewer than 250 employees and less than USD 10 million annual revenue**. Larger organisations need a paid Docker subscription.

> **Check:** if the PC belongs to a company above these limits, IT must confirm licensing before installing.

4. Part A: Installation IT / ADMINISTRATOR

These steps need local administrator rights. A standard user account cannot complete them.

1 Enable WSL 2

Open **PowerShell as Administrator** (Start → type *PowerShell* → right-click → *Run as administrator*) and run:

```
wsl --install --no-distribution
```

Restart the PC when it finishes. Then, again in PowerShell as Administrator, install the current WSL and its Linux kernel:

```
wsl --update
# If the Microsoft Store is blocked:  wsl --update --web-download
```

> **Windows 10:** skipping `wsl --update` makes Docker Desktop show "WSL needs updating" and the engine never starts. See **DEV-GUIDE-02**.

2 Install Docker Desktop

1. Download *Docker Desktop for Windows – AMD64* from `https://www.docker.com/products/docker-desktop/`
2. Run `Docker Desktop Installer.exe`.
3. On the configuration screen, tick **"Use WSL 2 instead of Hyper-V (recommended)"**.
4. Finish the installer and sign out of Windows if prompted.

3 Allow the developer account to use Docker without admin rights (all-users install only)

If Docker Desktop was installed for the current user only (it lives in `%LOCALAPPDATA%\Programs\DockerDesktop`), skip this step. The `docker-users` group isn't used.

```
# PowerShell as Administrator. Replace with the developer's DOMAIN\username
net localgroup docker-users "SCIENTER\sachithi" /add
```

4 Developer signs out and signs back in to Windows DEVELOPER

The new group membership only takes effect after signing in again.

> **Done:** from this point on, the developer does everything without administrator rights.

5. Part B: First-time configuration DEVELOPER

5.1 Limit Docker's memory and CPU

By default WSL 2 can use most of the PC's memory. Create the file `C:\Users\<username>\.wslconfig` (Notepad is fine; no admin needed):

```
[wsl2]
memory=6GB
processors=4
swap=2GB
```

Apply it:

```
wsl --shutdown
```

Then start Docker Desktop again.

> **Why 6 GB:** SQL Server needs at least 2 GB; the API, storefront and web server need about 1.5 GB together. This leaves the rest of a 16 GB PC for the IDE and browser.

5.2 Docker Desktop settings

Setting (Docker Desktop → Settings)	Value
General → Use the WSL 2 based engine	On
General → Start Docker Desktop when you sign in	Your choice (Off saves memory when not coding)
Resources → Advanced → Disk image location	`D:\DockerData` (drive with the most free space)
Resources → Proxies	Only if the company network uses a proxy (ask IT)
Software updates	Keep automatic update checks on

5.3 Verify the installation

```
docker --version
docker compose version
docker run --rm hello-world
```

> **Expected:** version numbers are printed, and the last command shows *"Hello from Docker!"*.

6. Part C: Running the project

6.1 First run

```
cd "D:\Projects\e com\spencer-denim-ecommerce"
git pull

# Create your local settings file from the template (never committed to git)
Copy-Item .env.example .env
# Open .env and set a strong SQL Server SA password:
# 8+ characters with upper case, lower case, a number and a symbol

# Build and start all services in the background
docker compose up -d --build

# Confirm every service shows "running" / "healthy"
docker compose ps
```

The first build downloads the base images and can take 5–15 minutes. Later starts take seconds.

6.2 Services and local addresses

Service	Purpose	Local address
`caddy`	Web server / reverse proxy in front of all services	`http://localhost`
`storefront`	Angular 21 customer website (server-side rendered)	`http://localhost`
`admin`	Angular 21 admin panel	`http://localhost/admin`
`api`	.NET 10 API; applies database migrations and seed data on start	`http://localhost/api` Docs: `/api/swagger`
`sqlserver`	SQL Server 2025 Express database	`localhost,1433` (this PC only)

6.3 Connecting to the database

Use **SQL Server Management Studio (SSMS)** or **VS Code with the "SQL Server (mssql)" extension**:

- Server: `localhost,1433`
- Authentication: SQL Login, with the user and password from your `.env` file
- Tick *Trust server certificate* (local development only)

6.4 Everyday commands

Command	What it does
`docker compose up -d`	Start all services (without rebuilding)
`docker compose ps`	Show service status
`docker compose logs -f api`	Follow the API logs (Ctrl+C stops following, not the service)
`docker compose restart api`	Restart one service
`docker compose up -d --build api`	Rebuild and restart one service after code changes
`docker compose down`	Stop everything. **Database data is kept.**
`docker compose down -v`	Stop everything **and delete the local database** (fresh start)

> **Careful:** `-v` permanently deletes the local database volume. Use it only when you want a clean database.

7. Part D: Recommended daily workflow

Rebuilding containers after every code change is slow. While coding, run **only the database in Docker** and run the API and Angular apps directly on the PC with hot reload:

```
# 1) Database only
docker compose up -d sqlserver

# 2) API with hot reload (terminal 1)
cd src\backend\SpencerDenim.Api
dotnet watch

# 3) Storefront or admin with hot reload (terminal 2)
cd src\frontend
npm install
npx ng serve storefront        # or: npx ng serve admin
```

When	Run
Writing code	Database in Docker, plus `dotnet watch` and `ng serve` on the PC
Running tests	`dotnet test`. Integration tests start their own temporary SQL Server container (Testcontainers) and remove it afterwards.
Before pushing to GitHub	Full stack: `docker compose up -d --build`, then check the site end to end
Finished for the day	`docker compose down` (data kept) and optionally quit Docker Desktop

8. Part E: Troubleshooting

Problem	Solution

"WSL needs updating" / engine never starts	WSL 2 kernel missing or outdated. Run `wsl --update` as Administrator, then `wsl --shutdown`, then Try Again. Full steps: **DEV-GUIDE-02**.
Docker Desktop stays on "Starting…"	Run `wsl --shutdown`, then reopen Docker Desktop. If it persists, restart Windows.
`permission denied` or "You are not allowed to use Docker"	Your account is not in `docker-users`. IT adds it (Part A, step 3), then sign out and back in.
"Virtualization support not detected"	IT enables virtualization (Intel VT-x / AMD-V) in the BIOS/UEFI.
Image downloads fail or time out	Company proxy/firewall. Set the proxy under Settings → Resources → Proxies, or ask IT to allow Docker Hub and mcr.microsoft.com.
Port 80 or 1433 is already in use	Another web server (IIS) or SQL Server is running. Stop it, or change the port in `.env`.
SQL Server container keeps restarting	Not enough memory, or a weak SA password. Check `.wslconfig` memory (≥ 6 GB) and the password rules in `.env`. View the reason with `docker compose logs sqlserver`.
PC becomes slow	Lower `memory` in `.wslconfig`, run `docker compose down` when not coding, and close unused containers.
Disk space running low	`docker system prune` removes unused images and build cache (your database volume is kept). Check usage with `docker system df`.
Code changes not visible in containers	Rebuild the service: `docker compose up -d --build <service>`

9. Security rules

- **Never commit** `.env` or any file with passwords, certificates, or PayPal/SMTP credentials. `.gitignore` blocks them, but check `git status` before committing.
- The local SQL Server port is bound to **this PC only**. On the production server the database port is **never** exposed to the internet.
- Use strong, unique passwords, even locally, and different passwords for local, staging and production.
- Keep Docker Desktop updated for security fixes.
- Pull base images only from official sources (Docker Official Images, mcr.microsoft.com).
- The project is pushed **only** to the personal repository `github.com/MASACHITH/spencer-denim-ecommerce` (see `docs/09-git-and-repository-rules.md`).

10. Quick reference

Task	Command
Check Docker is working	`docker run --rm hello-world`
Start everything	`docker compose up -d`
Start everything after code changes	`docker compose up -d --build`
Database only (daily coding)	`docker compose up -d sqlserver`
Service status	`docker compose ps`
Follow logs	`docker compose logs -f <service>`
Stop (keep data)	`docker compose down`
Stop and reset database	`docker compose down -v`
Restart WSL / fix a stuck Docker	`wsl --shutdown`
Free disk space	`docker system prune`

Document control: DEV-GUIDE-01 v1.1 (added wsl --update step and per-user install note), 17 September 2026, prepared by TU2SL for the Spencer Denim E-Commerce Platform. Source file: `docs/guides/docker-desktop-windows-setup-guide.html`.